

1 Process discovery Exercises

Exercise 1.1 The following ten event logs were extracted from an enterprise information system.

1. $L_1 = [5 \cdot \langle a, b, c, d \rangle, 8 \cdot \langle a, c, b, d \rangle, 9 \cdot \langle a, e, d \rangle]$
2. $L_2 = [\langle a, b, c, d, e, f, b, d, c, e, g \rangle, \langle a, b, d, c, e, g \rangle, \langle a, b, c, d, e, f, b, c, d, e, f, b, d, c, e, g \rangle]$
3. $L_3 = [45 \cdot \langle a, c, d \rangle, 42 \cdot \langle b, c, d \rangle, 38 \cdot \langle a, c, e \rangle, 22 \cdot \langle b, c, e \rangle]$
4. $L_4 = [2 \cdot \langle a, b, e, f \rangle, 3 \cdot \langle a, b, e, c, d, b, f \rangle, 2 \cdot \langle a, b, c, e, d, b, f \rangle, 4 \cdot \langle a, b, c, d, e, b, f \rangle, 3 \cdot \langle a, e, b, c, d, b, f \rangle]$
5. $L_5 = [\langle a, d, e, f, h \rangle, \langle a, e, d, f, h \rangle, \langle g, h \rangle, \langle a, b, c, d, f, h \rangle, \langle a, c, b, d, f, h \rangle, \langle a, b, d, c, f, h \rangle, \langle a, c, d, b, f, h \rangle]$
6. $L_6 = [\langle a, b, c, d, e, f, b, d, c, e, g \rangle, \langle a, b, d, c, e, g \rangle, \langle a, b, c, d, e, f, b, c, d, e, f, b, d, c, e, g \rangle]$
7. $L_7 = [\langle a, b, c, d, f \rangle, \langle a, c, b, d, f \rangle, \langle a, b, d, c, f \rangle, \langle a, c, d, b, f \rangle, \langle a, d, e, f \rangle, \langle a, e, d, f \rangle]$
8. $L_8 = [\langle a, b, c, d, e, f, b, d, c, e, g \rangle, \langle a, b, c, d, e, g \rangle, \langle h, e, g \rangle]$
9. $L_9 = [\langle a, d, f, h \rangle, \langle a, c, e, g, c, e, h \rangle, \langle b, f, g, d, f, h \rangle]$
10. $L_{10} = [\langle a, b, d, e, g, i, j \rangle, \langle a, b, e, d, h, i, j \rangle, \langle a, b, f, d, g, i, j \rangle, \langle a, b, f, g, d, i, j \rangle, \langle a, b, g, d, f, i, j \rangle, \langle a, b, d, f, h, i, c, b, e, h, d, i, j \rangle, \langle a, b, g, f, d, i, c, b, h, f, d, i, j \rangle, \langle a, b, h, e, d, i, c, b, d, g, e, i, j \rangle]$
11. $L_{11} = [\langle a, c, d, f \rangle, \langle a, c, d, f \rangle, \langle b, e, f \rangle, \langle a, d, c, f \rangle, \langle b, e, f \rangle, \langle a, d, c, f \rangle, \langle b, e, f \rangle]$
12. $L_{12} = [\langle a, c, d, f \rangle, \langle a, c, d, g \rangle, \langle b, e, f \rangle, \langle a, d, c, g \rangle, \langle b, e, g \rangle, \langle a, d, c, f \rangle, \langle b, e, f \rangle]$
13. $L_{13} = [\langle f, b, g, a, d \rangle, \langle f, c, a, h, i, e \rangle, \langle f, c, h, a, i, e \rangle, \langle f, b, a, g, d \rangle, \langle f, c, h, i, a, e \rangle]$
14. $L_{14} = [\langle d, a, g, f, b, c \rangle, \langle d, a, h, f, b, i, a, g, f, b, c \rangle, \langle e, a, f, h, b, i, a, f, h, b, c \rangle, \langle e, a, f, g, b, c \rangle, \langle e, a, h, f, b, c \rangle]$

- Derive the $\rightarrow_L$ relation.
- Use the eight steps of the α -algorithm to construct the corresponding Petri net and draw the Petri net (delivering all of the intermediate results is not necessary, only the resulting Petri net is required).
- If possible, give a trace possible according to the discovered model but not (yet) observed in the log.
- Try the Heuristics and Inductive Miner (WARNING: no solution provided for this, but use the ProM tool introduced in the lab to see the solutions.).

Solutions

Solution 1.1

1.
 - $\rightarrow_{L_1} = \{(a, b), (a, c), (a, e), (b, d), (c, d), (e, d)\}$
 - See Figure 1.
 - There is no such trace.

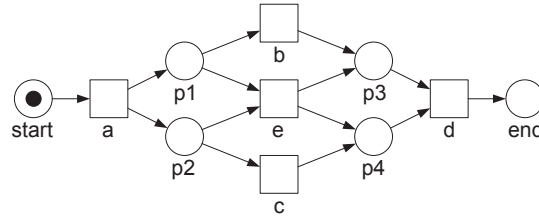


Figure 1: The process model discovered from L_1 .

2.
 - $\rightarrow_{L_2} = \{(a, b), (b, c), (b, d), (c, e), (d, e), (e, f), (e, g), (f, b)\}$
 - See Figure 2.
 - For example, $\langle a, b, d, c, e, f, b, c, d, e, g \rangle$.

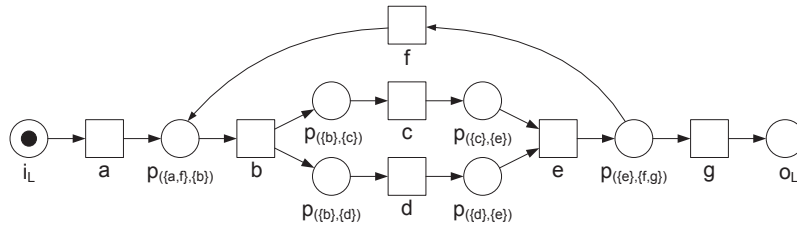


Figure 2: The process model discovered from L_2 .

3.
 - $\rightarrow_{L_3} = \{(a, c), (b, c), (c, d), (c, e)\}$
 - See Figure 3.
 - There is no such trace.

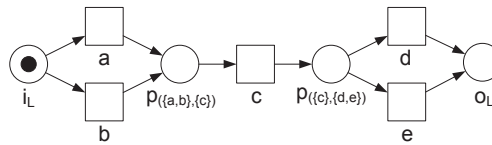


Figure 3: The process model discovered from L_3 .

4.
 - $\rightarrow_{L_4} = \{(a, b), (a, e), (b, c), (b, f), (c, d), (d, b), (e, f)\}$
 - See Figure 4.
 - For example, $\langle a, e, b, f \rangle$.

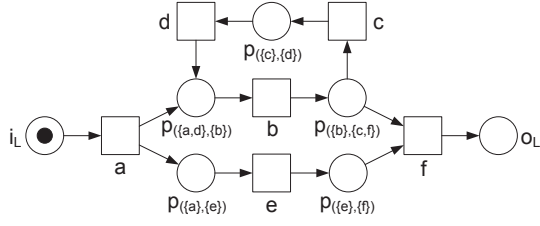


Figure 4: The process model discovered from L_4 .

5.
 - $\rightarrow_{L_5} = \{(a, b), (a, c), (a, d), (a, e), (b, f), (c, f), (d, f), (e, f), (f, h), (g, h)\}$
 - See Figure 5.
 - $\langle a, d, b, c, f, h \rangle$ and $\langle a, d, c, b, f, h \rangle$.

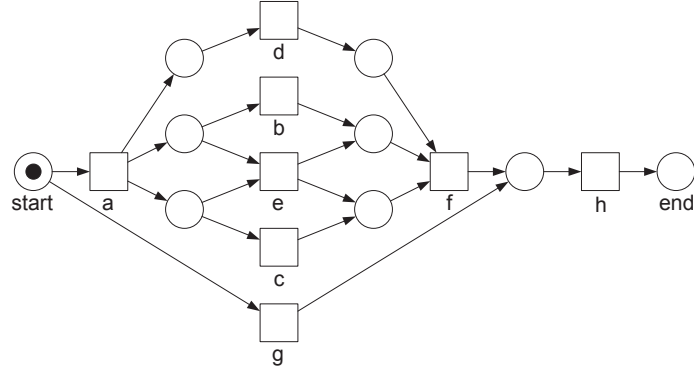


Figure 5: The process model discovered from L_5 .

6.
 - $\rightarrow_{L_6} = \{(a, b), (b, c), (b, d), (c, e), (d, e), (e, f), (e, g), (f, b)\}$
 - See Figure 6.
 - For example, $\langle a, b, c, d, e, g \rangle$.

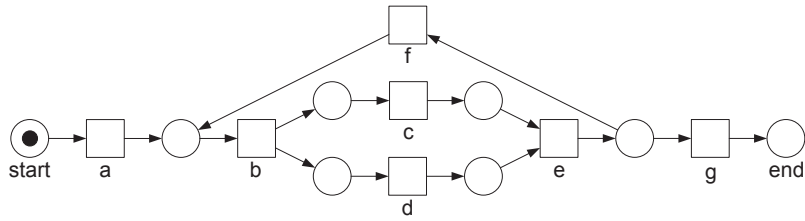


Figure 6: The process model discovered from L_6 .

7.
 - $\rightarrow_{L_7} = \{(a, b), (a, c), (a, d), (a, e), (b, f), (c, f), (d, f), (e, f)\}$
 - See Figure 7.
 - For example, $\langle a, d, b, c, f \rangle$ and $\langle a, d, c, b, f \rangle$.

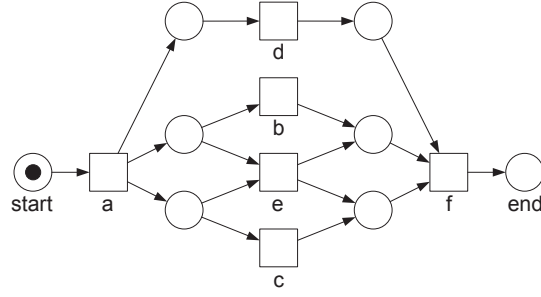


Figure 7: The process model discovered from L_7 .

- 8.
- $\rightarrow_{L_8} = \{(a, b), (b, c), (b, d), (c, e), (d, e), (e, f), (e, g), (f, b), (h, e)\}$
 - See Figure 8.
 - For example, $\langle h, e, f, b, c, d, e, g \rangle$.

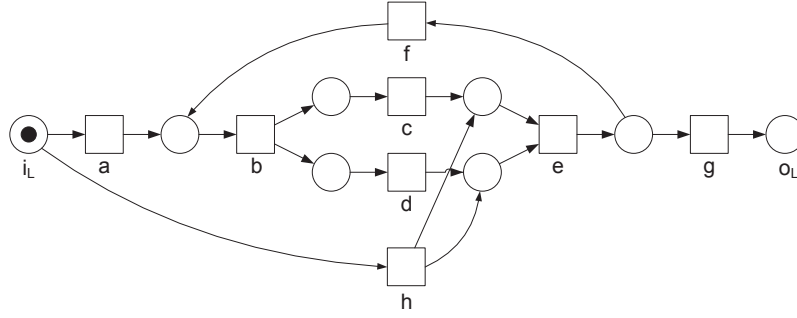


Figure 8: The process model discovered from L_8 .

- 9.
- $\rightarrow_{L_9} = \{(a, c), (a, d), (b, f), (c, e), (d, f), (e, g), (e, h), (f, g), (f, h), (g, c), (g, d)\}$
 - See Figure 9.
 - For example, $\langle b, f, g, c, e, h \rangle$.

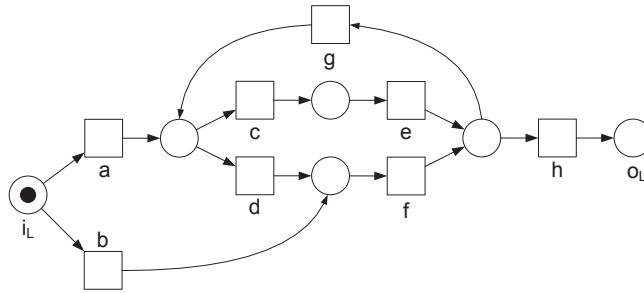


Figure 9: The process model discovered from L_9 .

- 10.
- $\rightarrow_{L_{10}} = \{(a, b), (b, d), (b, e), (b, f), (b, g), (b, h), (c, b), (d, i), (e, i), (f, i), (h, i), (g, i), (i, c), (i, j)\}$
 - See Figure 10.
 - For example, $\langle a, b, d, h, e, i, j \rangle$.

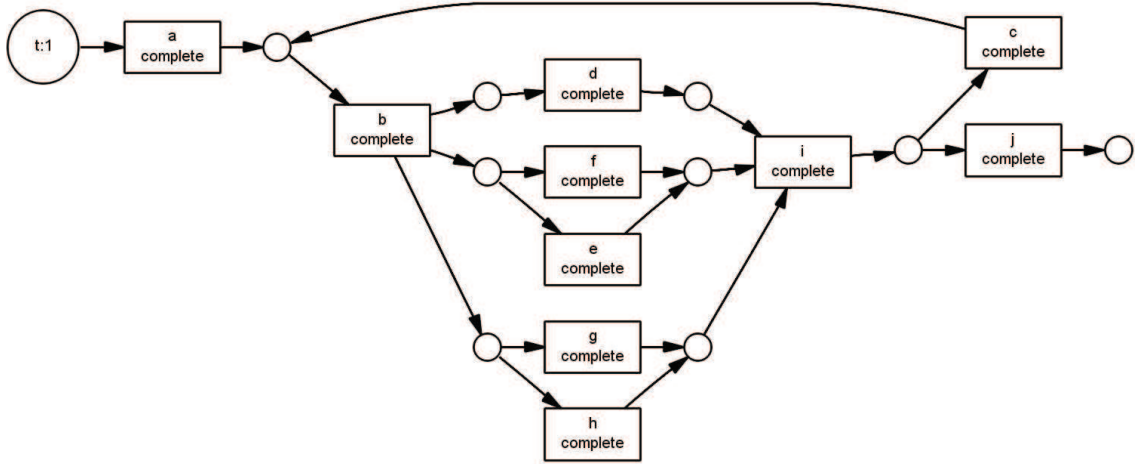


Figure 10: The process model discovered from L_{10} .

11. • $\rightarrow_{L_{11}} = \{(a, c), (a, d), (b, e), (c, f), (d, f), (e, f)\}$
 • See Figure 11.
 • There is no such trace.

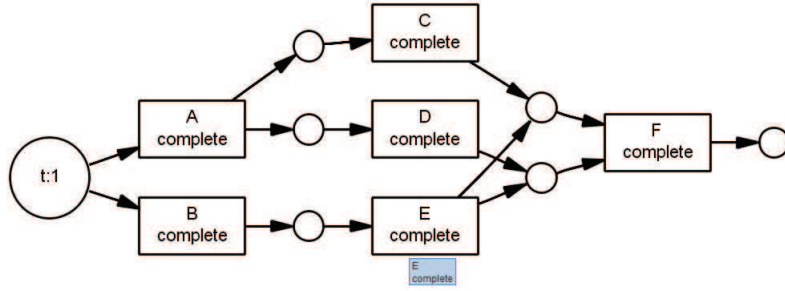


Figure 11: The process model discovered from L_{11} .

12. • $\rightarrow_{L_{12}} = \{(a, c), (a, d), (b, e), (c, f), (c, g), (d, f), (d, g), (e, f), (e, g)\}$
 • See Figure 12.
 • There is no such trace.

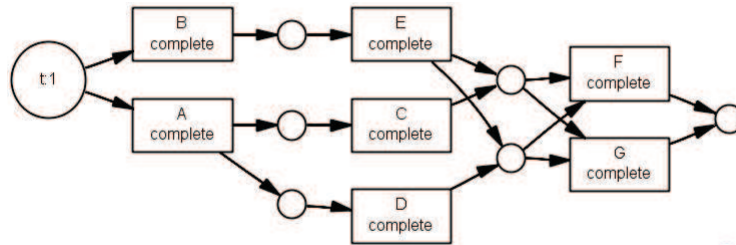


Figure 12: The process model discovered from L_{12} .

13. • $\rightarrow_{L_{13}} = \{(f, b), (f, c), (b, g), (b, a), (c, a), (c, h), (g, d), (a, d), (h, i), (a, e), (i, e)\}$
 • See Figure 13.

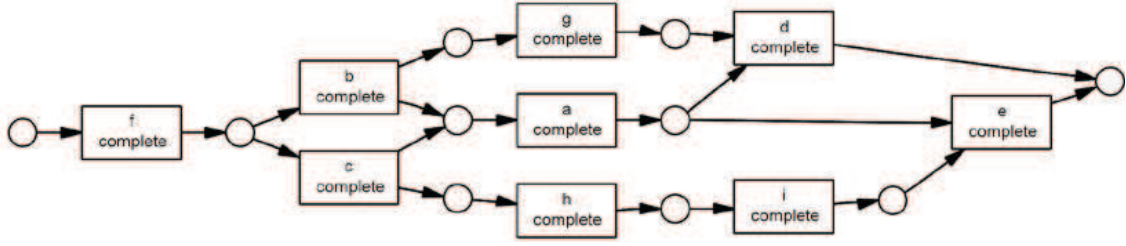


Figure 13: The process model discovered from L_{13} .

- There is no such trace.
14. • $\rightarrow_{L_{14}} = \{(d, a), (e, a), (a, g), (a, h), (a, f), (g, b), (h, b), (f, b), (b, i), (b, c), (i, a)\}$
- See Figure 14.
 - For example $\langle d, a, g, f, b, i, a, g, f, b, c \rangle$.

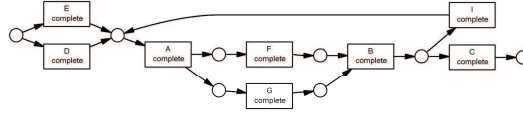


Figure 14: The process model discovered from L_{14} .